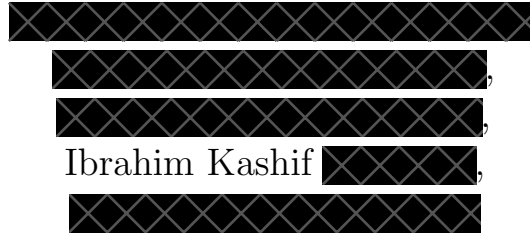


On the Properties and Applications of Quaternions



1st May 2026

Abstract

This paper starts with a historical overview of quaternions and explores their algebraic properties. Moreover, we discuss two applications of quaternions, namely a representation of rotations in $\mathbb{R}^3$ and a proof of Lagrange's four-square theorem. Finally, we explore the octonions and introduce the Cayley-Dickson construction.

1 Introduction

Quaternions can be thought of as four-dimensional numbers. They are an extension of complex numbers, which are two-dimensional. They were first defined by William Rowan Hamilton when he studied rotations in three-dimensional space. We first give an overview of how Hamilton invented the quaternions, using the description in [33, pp. 228–229].

After Hamilton gave the first formal definition of complex numbers in [17, p. 408] as a couple of real numbers, which were used to study rotations in $\mathbb{R}^2$, he tried to use triplets of real numbers to study rotations in $\mathbb{R}^3$. Using a similar notation to complex numbers, he wrote the triplets as $a + bi + cj$. However, he could not define a well-behaved multiplication of triplets. He required the multiplication to satisfy two conditions. First, it was possible to multiply the triplets term by term. This means that given two triplets $a_1 + b_1i + c_1j$ and $a_2 + b_2i + c_2j$, we can define i^2 , j^2 , ij and ji so that $(a_1 + b_1i + c_1j)(a_2 + b_2i + c_2j) = a_1a_2 + (a_1b_2 + b_1a_2)i + (a_1c_2 + c_1a_2)j + b_1b_2i^2 +$

$c_1c_2j^2 + b_1c_2ij + c_1b_2ji$ matches the result of the definition of multiplication. Second, the norm should be multiplicative, which means $\|uv\| = \|u\| \cdot \|v\|$ for any triplets u and v . This structure is now known as a normed division algebra. In fact, it was later proved that three-dimensional normed division algebra does not exist. Failing to define a multiplication on triplets, Hamilton moved to four dimensions and invented the quaternions eventually. We will define the quaternions formally, explain their basic properties, and explore some algebraic structures formed by quaternions in Section 2.

Quaternions are used in many areas of mathematics, such as proving Lagrange’s four-square theorem in number theory. Mathematicians in the 17th and 18th centuries asked the problem of how we can classify integers by the least amount of squares in a sum. Finally in 1770, Joseph-Louis Lagrange proved that every positive integer can be expressed as a sum of four squares, which is known as Lagrange’s four-square theorem [13]. This theorem has elementary proofs based on arithmetic reasoning. However, these methods are usually computationally long and hard to understand as they use the method of infinite descent, such as in [19, pp. 399–400]. They also fail to make apparent why sums of four squares behave well under multiplication. Quaternions provide an alternative proof of this theorem, which also make the behaviour of the multiplication easily seen and simplify the computation. We explain the proof of Lagrange’s four-square theorem using quaternions in detail in Section 3.

Furthermore, there are many real-world applications of quaternions. For example, they can be used to represent rotations in $\mathbb{R}^3$, as this is the reason why Hamilton invented the quaternions. Although there are many other ways to represent rotations in $\mathbb{R}^3$, such as using orthonormal 3×3 matrices and Euler angles as suggested in [20, p. 556], these methods can sometimes cause problems. Euler angles lead to a problem called gimbal lock, which we discuss in Section 4. Orthonormal 3×3 matrices can be affected by numerical drift, meaning that the matrix may no longer be a valid rotation matrix, according to [20, p. 556]. However, quaternions provide a clear and compact way to represent rotations, avoiding these issues. In Section 4, we explain how quaternions represent rotations and how they solve the problem of gimbal lock.

Recall that the quaternions is an extension of complex numbers. However, quaternions can be further extended to octonions, an eight-dimensional number, which can again be extended to sedenions, a 16-dimensional number. This construction process, called the Cayley-Dickson construction, can be continued forever. However, as we increase the dimension of the numbers, some properties of $\mathbb{R}$ are lost. In Section 5, we define the octonions formally and state the Cayley-Dickson construction. We also explore how

the properties are lost throughout the construction.

2 Definitions and properties

In this section, we use the definition of quaternions given in [18, pp. 1–2]. We introduce another equivalent definition in Section 5.2.

Definition 2.1. A quaternion is an expression of the form $q = a + bi + cj + dk$, where $a, b, c, d \in \mathbb{R}$ are called the constituents and i, j, k are symbols of three imaginary quantities, which we call imaginary units.

Remark. We can treat a vector $v = \alpha i + \beta j + \gamma k \in \mathbb{R}^3$ as a quaternion $v = 0 + \alpha i + \beta j + \gamma k \in \mathbb{H}$.

The set of all quaternions is denoted by $\mathbb{H}$, while the set of all non-zero quaternions is denoted by $\mathbb{H}^*$.

Definition 2.2. The addition and subtraction of two quaternions $p = a_1 + b_1i + c_1j + d_1k$ and $q = a_2 + b_2i + c_2j + d_2k$ are defined as

$$p \pm q = (a_1 \pm a_2) + (b_1 \pm b_2)i + (c_1 \pm c_2)j + (d_1 \pm d_2)k.$$

Definition 2.3. The product of two quaternions, $p = a_1 + b_1i + c_1j + d_1k$ and $q = a_2 + b_2i + c_2j + d_2k$, can be evaluated by expanding the brackets and using the following rules:

$$\begin{aligned} i^2 &= j^2 = k^2 = -1, \\ ij &= k, \quad jk = i, \quad ki = j, \\ ji &= -k, \quad kj = -i, \quad ik = -j. \end{aligned}$$

Example 2.4 (Evaluate $(4 + 2i - 3j + 5k)(2 - i + 4j + 2k)$).

$$\begin{aligned} &(4 + 2i - 3j + 5k)(2 - i + 4j + 2k) \\ &= 8 - 4i + 16j + 8k + 4i - 2i^2 + 8ij + 4ik \\ &\quad - 6j + 3ji - 12j^2 - 6jk + 10k - 5ki + 20kj + 10k^2 \\ &= 8 - 4i + 16j + 8k + 4i - 2(-1) + 8k + 4(-j) \\ &\quad - 6j + 3(-k) - 12(-1) - 6i + 10k - 5j + 20(-i) + 10(-1) \\ &= 8 - 4i + 16j + 8k + 4i + 2 + 8k - 4j \\ &\quad - 6j - 3k + 12 - 6i + 10k - 5j - 20i - 10 \\ &= 12 - 26i + j + 23k. \end{aligned}$$

Theorem 2.5 (Associative, commutative and distributive laws, [16], p. 1]).
Let $p, q, r \in \mathbb{H}$. Then, we have

1. $(p + q) + r = p + (q + r)$,
2. $p + q = q + p$,
3. $(pq)r = p(qr)$,
4. $p(q + r) = pq + pr$,
5. $(q + r)p = qp + rp$.

However, multiplication of quaternions is not commutative in general. For example, $ij = k$ but $ji = -k$.

If $q \in \mathbb{H}$, then it is easy to see that $q + 0 = q$, $q - q = 0$ and $q1 = 1q = q$. These properties, together with Theorem 2.5, imply that $\mathbb{H}$ is a ring. In fact, $\mathbb{H}$ is also a division ring, which is defined as follows.

Definition 2.6 ([22], Definition 2.3]). A ring R is a division ring if every element r in R , except the additive identity 0_R , has a multiplicative inverse r^{-1} in R satisfying $rr^{-1} = r^{-1}r = 1_R$, where 1_R is the multiplicative identity.

It is obvious that $\mathbb{H}$ is a division ring after we define the division of non-zero quaternions later in this section. Similar to the complex numbers, we can define the conjugate and the norm of a quaternion.

Definition 2.7 ([30], Section 8.6]). The conjugate of a quaternion $q = a + bi + cj + dk$ is defined as $a - bi - cj - dk$ and is denoted by $\bar{q}$.

Definition 2.8. The norm of a quaternion $q = a + bi + cj + dk$ is defined as $\sqrt{a^2 + b^2 + c^2 + d^2}$ and is denoted by $\|q\|$.

Remark. To say that $\|\cdot\|$ is a norm, we should verify that it indeed satisfies the axioms of a norm. We formally define what a norm is and prove that it is indeed a norm later in Definition 2.15 and Theorem 2.17 respectively.

Lemma 2.9 ([30], p. 149]). Let $p, q \in \mathbb{H}$. Then the following properties hold.

1. $\overline{p + q} = \bar{p} + \bar{q}$.
2. $\overline{pq} = \bar{q}\bar{p}$.
3. $q\bar{q} = \bar{q}q = \|q\|^2$.

Definition 2.10. A unit quaternion is a quaternion q such that $\|q\| = 1$.

The set of all unit quaternions is denoted by $\mathbb{H}^1$.

From Lemma 2.9, we have $q\bar{q} = \bar{q}q = \|q\|^2$. Furthermore, we have $\|q\| = 0$ if and only if $q = 0$. Therefore, if $q \neq 0$, then we can divide the equation by $\|q\|^2$ to get

$$q \frac{\bar{q}}{\|q\|^2} = \frac{\bar{q}}{\|q\|^2} q = 1.$$

Hence, the multiplicative inverse of a non-zero quaternion q is $q^{-1} = \bar{q}/\|q\|^2$. The existence of multiplicative inverse of non-zero quaternions also implies that $\mathbb{H}$ is a division ring. Finally, the division of quaternions p and q , where $q \neq 0$, is defined as $p/q = pq^{-1} = p(\bar{q}/\|q\|^2)$.

The associative law in Theorem 2.5, with 1 as the multiplicative identity and the existence of multiplicative inverses for non-zero quaternions, imply that $\mathbb{H}^*$ forms a group under multiplication. This fact is useful when we discuss representations of quaternions in Section 4. Furthermore, the following lemma follows from the properties of a group.

Lemma 2.11 ([21, Theorem 1.1]). *Let $p, q \in \mathbb{H}^*$. Then $(pq)^{-1} = q^{-1}p^{-1}$.*

Another algebraic structure that $\mathbb{H}$ forms is an $\mathbb{R}$ -vector space. Nevertheless, $\mathbb{H}$ satisfies more properties than required by a vector space and it in fact forms a normed division algebra over $\mathbb{R}$. We first define what algebra, division algebra, and normed vector space are. Finally, we define what is a normed division algebra.

Definition 2.12 ([27, p. 1]). Let V be a $\mathbb{R}$ -vector space, with operations $+$: $V \times V \rightarrow V$ and $\cdot$: $\mathbb{R} \times V \rightarrow V$. Suppose we have an operation $\times$: $V \times V \rightarrow V$ such that for all $u, v, w \in V$ and $\lambda, \mu \in \mathbb{R}$, we have

$$(A1) \quad (u + v) \times w = u \times w + v \times w,$$

$$(A2) \quad u \times (v + w) = u \times v + u \times w,$$

$$(A3) \quad (\lambda u) \times (\mu v) = (\lambda\mu)(u \times v).$$

Then, we say that V is a $\mathbb{R}$ -algebra.

Remark. The dimension of an $\mathbb{R}$ -algebra V is the same as the dimension of V as an $\mathbb{R}$ -vector space. Also, we may write uv for $u \times v$.

Definition 2.13 ([25, Section 1]). A division algebra over $\mathbb{R}$ is a finite-dimensional $\mathbb{R}$ -algebra such that it contains an identity for the operation $\times$ and it has no zero divisors, which means that for any non-zero u and v , we have $uv \neq 0$.

Example 2.14 ([3, p. 3]). Under the usual additions and multiplications, $\mathbb{R}$, $\mathbb{C}$, and $\mathbb{H}$ are division algebras over $\mathbb{R}$.

Definition 2.15 ([1, p. 194]). Let V be a $\mathbb{R}$ -vector space. Let $\|\cdot\| : V \rightarrow \mathbb{R}$ be a function satisfying the following axioms.

(N1) For any $v \in V$, $\|v\| \geq 0$.

(N2) For any $v \in V$, $\|v\| = 0$ if and only if $v = 0$.

(N3) For any $v \in V$ and $\lambda \in \mathbb{R}$, $\|\lambda v\| = |\lambda| \|v\|$.

(N4) For any $u, v \in V$, $\|u + v\| \leq \|u\| + \|v\|$.

Then, we say that $(V, \|\cdot\|)$ is a normed vector space.

Definition 2.16 ([3, p. 3]). Let $(V, \|\cdot\|)$ be a normed vector space over $\mathbb{R}$ such that V is also a division algebra over $\mathbb{R}$. We say that V is a normed division algebra over $\mathbb{R}$ if $\|uv\| = \|u\| \cdot \|v\|$ for any $u, v \in V$.

Theorem 2.17. *Under the usual addition and multiplication, $\mathbb{H}$ is a normed division algebra over $\mathbb{R}$.*

Proof. We first prove that $(\mathbb{H}, \|\cdot\|)$ is a normed vector space. (N1) and (N2) follow from the properties of real numbers. Let $\lambda \in \mathbb{R}$ and $p, q \in \mathbb{H}$, where $p = a_1 + b_1i + c_1j + d_1k$ and $q = a_2 + b_2i + c_2j + d_2k$. Then we have

$$\begin{aligned} \|\lambda p\| &= \|\lambda a_1 + \lambda b_1i + \lambda c_1j + \lambda d_1k\| = \sqrt{(\lambda a_1)^2 + (\lambda b_1)^2 + (\lambda c_1)^2 + (\lambda d_1)^2} \\ &= \sqrt{\lambda^2(a_1^2 + b_1^2 + c_1^2 + d_1^2)} = \sqrt{\lambda^2} \sqrt{a_1^2 + b_1^2 + c_1^2 + d_1^2} = |\lambda| \|p\|. \end{aligned}$$

Therefore, (N3) is satisfied. Moreover,

$$\begin{aligned} \|p + q\|^2 &= \|(a_1 + a_2) + (b_1 + b_2)i + (c_1 + c_2)j + (d_1 + d_2)k\|^2 \\ &= (a_1 + a_2)^2 + (b_1 + b_2)^2 + (c_1 + c_2)^2 + (d_1 + d_2)^2 \\ &= (a_1^2 + b_1^2 + c_1^2 + d_1^2) + (a_2^2 + b_2^2 + c_2^2 + d_2^2) + 2(a_1a_2 + b_1b_2 + c_1c_2 + d_1d_2) \\ &= \|p\|^2 + \|q\|^2 + 2(a_1a_2 + b_1b_2 + c_1c_2 + d_1d_2). \end{aligned}$$

By the Cauchy-Schwarz inequality in [14, Theorem 1.2], $(a_1a_2 + b_1b_2 + c_1c_2 + d_1d_2)^2 \leq (a_1^2 + b_1^2 + c_1^2 + d_1^2)(a_2^2 + b_2^2 + c_2^2 + d_2^2)$. Taking square roots on both sides, we have $a_1a_2 + b_1b_2 + c_1c_2 + d_1d_2 \leq \sqrt{a_1^2 + b_1^2 + c_1^2 + d_1^2} \sqrt{a_2^2 + b_2^2 + c_2^2 + d_2^2} = \|p\| \cdot \|q\|$. Therefore,

$$\begin{aligned} \|p + q\|^2 &= \|p\|^2 + \|q\|^2 + 2(a_1a_2 + b_1b_2 + c_1c_2 + d_1d_2) \\ &\leq \|p\|^2 + \|q\|^2 + 2\|p\| \cdot \|q\| \\ &= (\|p\| + \|q\|)^2. \end{aligned}$$

Taking square roots on both sides, we get $\|p + q\| \leq \|p\| + \|q\|$. Hence, (N4) is also satisfied. Thus, $(\mathbb{H}, \|\cdot\|)$ is a normed vector space.

To show that $\mathbb{H}$ is a normed division algebra over $\mathbb{R}$, we also need to show that the norm is multiplicative. We note that $\|pq\| = \sqrt{(pq)(\overline{pq})} = \sqrt{(pq)(\overline{q}\overline{p})} = \sqrt{p(q\overline{q})\overline{p}} = \sqrt{p\|q\|^2\overline{p}} = \|q\|\sqrt{p\overline{p}} = \|p\| \cdot \|q\|$. Therefore, $\mathbb{H}$ is indeed a normed division algebra over $\mathbb{R}$. $\square$

3 Lagrange's four-square theorem

After establishing the definition and properties of quaternions, we can present a proof of Lagrange's four-square theorem, which is based on a subset of quaternions called the Hurwitz integers.

Definition 3.1 ([10, Definition 2.2]). The set of Hurwitz integers is defined as

$$\mathbb{H}_{\text{Hurwitz}} = \{a+bi+cj+dk \mid a, b, c, d \in \mathbb{Z}\} \cup \left\{a+bi+cj+dk \mid a, b, c, d \in \mathbb{Z} + \frac{1}{2}\right\},$$

where $\mathbb{Z} + \frac{1}{2} = \{n + \frac{1}{2} \mid n \in \mathbb{Z}\}$.

Hurwitz integers is a key component in proving Lagrange's four-square theorem because they have valuable behaviours and properties. For example, the squared norm of a Hurwitz integer is always an integer. To prove this fact, we need the following lemma.

Lemma 3.2. *Let n be an odd integer. Then $n^2 \equiv 1 \pmod{4}$.*

Proof. Since n is odd, there exists $m \in \mathbb{Z}$ such that $n = 2k + 1$. Hence, $n^2 = (2m + 1)^2 = 4m^2 + 4m + 1 = 4(m^2 + m) + 1 \equiv 1 \pmod{4}$. $\square$

Theorem 3.3 ([30, p. 148]). *For all $q \in \mathbb{H}_{\text{Hurwitz}}$, we have $\|q\|^2 \in \mathbb{Z}$.*

Proof. Let $q = a + bi + cj + dk$. We consider two cases. The first case is $a, b, c, d \in \mathbb{Z}$. Then, we have $\|q\|^2 = a^2 + b^2 + c^2 + d^2$. Since $a^2, b^2, c^2, d^2 \in \mathbb{Z}$, we have $\|q\|^2 \in \mathbb{Z}$.

For the second case, where $a, b, c, d \in \mathbb{Z} + \frac{1}{2}$, note that we can rewrite $q = a + bi + cj + dk$ as $(\alpha + \beta i + \gamma j + \delta k)/2$, where $\alpha, \beta, \gamma, \delta$ are odd numbers. By Lemma 3.2, we have $\alpha^2, \beta^2, \gamma^2, \delta^2 \equiv 1 \pmod{4}$. Also, we have $\|q\|^2 = (\alpha^2 + \beta^2 + \gamma^2 + \delta^2)/4$. Since $\alpha^2 + \beta^2 + \gamma^2 + \delta^2 \equiv 1 + 1 + 1 + 1 = 4 \equiv 0 \pmod{4}$, $\alpha^2 + \beta^2 + \gamma^2 + \delta^2$ is divisible by 4. Hence, $\|q\|^2 \in \mathbb{Z}$.

Therefore, in both cases, we have $\|q\|^2 \in \mathbb{Z}$. $\square$

Theorem 3.4 (The four square identity, [30, p. 144]). *Let $a_1, b_1, c_1, d_1, a_2, b_2, c_2, d_2$ be integers. Then $(a_1^2 + b_1^2 + c_1^2 + d_1^2)(a_2^2 + b_2^2 + c_2^2 + d_2^2)$ is a sum of four integer squares.*

Proof. Define two quaternions $p = a_1 + b_1i + c_1j + d_1k$ and $q = a_2 + b_2i + c_2j + d_2k$. Note that $a_1^2 + b_1^2 + c_1^2 + d_1^2 = \|p\|^2$ and $a_2^2 + b_2^2 + c_2^2 + d_2^2 = \|q\|^2$. So we have $(a_1^2 + b_1^2 + c_1^2 + d_1^2)(a_2^2 + b_2^2 + c_2^2 + d_2^2) = \|p\|^2\|q\|^2 = \|pq\|^2$ since the norm is multiplicative. If we expand the brackets in $pq = (a_1 + b_1i + c_1j + d_1k)(a_2 + b_2i + c_2j + d_2k)$, we can show that all the constituents of pq are integers. This implies $pq = a_3 + b_3i + c_3j + d_3k$ for some $a_3, b_3, c_3, d_3 \in \mathbb{Z}$. Hence, $(a_1^2 + b_1^2 + c_1^2 + d_1^2)(a_2^2 + b_2^2 + c_2^2 + d_2^2) = \|a_3 + b_3i + c_3j + d_3k\|^2 = a_3^2 + b_3^2 + c_3^2 + d_3^2$. $\square$

The next corollary follows from Theorem 3.4 by induction.

Corollary 3.5. *Let $n \in \mathbb{N}$. Suppose $a_1, b_1, c_1, d_1, a_2, b_2, c_2, d_2, \dots, a_n, b_n, c_n, d_n \in \mathbb{Z}$. Then $(a_1^2 + b_1^2 + c_1^2 + d_1^2)(a_2^2 + b_2^2 + c_2^2 + d_2^2) \dots (a_n^2 + b_n^2 + c_n^2 + d_n^2)$ is a sum of four integer squares.*

As a result of Corollary 3.5, it suffices to prove that Lagrange's four-square theorem holds for all prime numbers, because any composite number can be factorised as a product of prime numbers and the theorem holds for 1 as $1 = 1^2 + 0^2 + 0^2 + 0^2$. Next, we define a special type of Hurwitz integers.

Definition 3.6 ([26, p. 4]). A Hurwitz prime is a Hurwitz integer q such that

1. $q \neq 0$,
2. $\|q\| \neq 1$, and
3. q cannot be written as a product of non-unit Hurwitz integers.

The following theorem provides a quick way to determine whether a given Hurwitz integer is a Hurwitz prime.

Theorem 3.7 ([26, Lemma 2.8]). *Let q be a Hurwitz integer. Then q is a Hurwitz prime if and only if $\|q\|^2$ is a prime number.*

Example 3.8 (Verify that the Hurwitz integers $p = 13 + 6i - 3j - 3k$ and $q = -\frac{7}{2} + \frac{3}{2}i - \frac{3}{2}j + \frac{1}{2}k$ are Hurwitz primes). Note that

$$\|p\|^2 = 13^2 + 6^2 + 3^2 + 3^2 = 223$$

and

$$\|q\|^2 = \left(\frac{7}{2}\right)^2 + \left(\frac{3}{2}\right)^2 + \left(\frac{3}{2}\right)^2 + \left(\frac{1}{2}\right)^2 = \frac{68}{4} = 17.$$

Since 223 and 17 are prime numbers, p and q are indeed Hurwitz primes.

As suggested by the names, it is natural to ask whether prime numbers are Hurwitz primes. To answer this question, we need to define the greatest common divisor (gcd) of two Hurwitz integers. However, since multiplication of quaternions is not commutative in general, we need to distinguish between right and left divisors, and thus between right and left gcd.

Definition 3.9 ([30, p. 151]). Let p and q be Hurwitz integers. We say that p is a right divisor of q if $q = rp$ for some $r \in \mathbb{H}_{\text{Hurwitz}}$.

The right gcd of two Hurwitz integers is defined analogously to the Euclidean algorithm for integers. We first recall the Euclidean algorithm.

Theorem 3.10 (Euclidean algorithm, [31, pp. 32–33]). Let $n, m \in \mathbb{Z}$. If one of them is zero, then the other is the greatest common divisor. If both are non-zero, then we can divide n by m to get a remainder r_1 , and write $n = q_1m + r_1$, for some $q_1 \in \mathbb{Z}$. If $r_1 \neq 0$, then we divide m by r_1 to get another remainder r_2 , and write $m = q_2r_1 + r_2$, for some $q_2 \in \mathbb{Z}$. If $r_2 \neq 0$, then we divide r_1 by r_2 to get the next remainder r_3 , and write $r_1 = q_3r_2 + r_3$, for some $q_3 \in \mathbb{Z}$. Repeat this process until we get a remainder of zero. The final non-zero remainder is $\gcd(n, m)$.

Definition 3.11 ([30, pp. 151–152]). Let $p, q \in \mathbb{H}_{\text{Hurwitz}}$. Note that if p and q have a common right divisor z , then $p = rz$ and $q = sz$ for some $r, s \in \mathbb{H}_{\text{Hurwitz}}$. Hence, when we divide p by q , we have

$$r_1 = p - \mu q = rz - \mu sz = (r - \mu s)z,$$

where r_1 is the remainder. This shows that z is a right divisor of r_1 as well. Hence, if we always divide on the right in the Euclidean algorithm, we get the right gcd for p and q . We denote this as $\text{rgcd}(p, q)$. Then, by inspection of the terms made in the Euclidean algorithm, we get that $\text{rgcd}(p, q) = \alpha p + \beta q$ for some $\alpha, \beta \in \mathbb{H}_{\text{Hurwitz}}$.

Definition 3.11 allows us to prove the following theorem about Hurwitz primes.

Theorem 3.12 ([30, p. 152]). If $x \in \mathbb{R}$ is a Hurwitz prime and x divides pq , where $p, q \in \mathbb{H}_{\text{Hurwitz}}$, then x divides p or x divides q .

Proof. Assume that x does not divide p . Then $1 = \text{rgcd}(x, p) = \alpha x + \beta p$, for some $\alpha, \beta \in \mathbb{H}_{\text{Hurwitz}}$. Thus, $q = \alpha xq + \beta pq$. Since x divides pq by assumption, x divides βpq . So x divides both αxq and βpq . Hence, x divides $\alpha xq + \beta pq = q$.

We proved that if x does not divide p , then x divides q . This is equivalent to x divides p or x divides q . $\square$

Lemma 3.13 ([30, p. 153]). *Given an odd prime p , there exist $q, r \in \mathbb{Z}$ such that p divides $1 + q^2 + r^2$.*

With Lemma 3.13, we can answer the question of whether odd prime integers are Hurwitz primes.

Lemma 3.14 ([30, p. 153]). *Any odd prime is not a Hurwitz prime.*

Proof. Let p be an odd prime. By Lemma 3.13, p divides $1 + q^2 + r^2$ for some $q, r \in \mathbb{Z}$. Note that we can factorise $1 + q^2 + r^2$ into Hurwitz integers as

$$1 + q^2 + r^2 = (1 + qi + rj)(1 - qi - rj).$$

We know that p divides $1 + q^2 + r^2$ but, both $\frac{1+qi+ri}{p}$ and $\frac{1-qi-rj}{p}$ are not Hurwitz integers, hence p does not divide $1 + qi + rj$ and $1 - qi - rj$. Using the contrapositive of Theorem 3.12, we can conclude that p is not a Hurwitz prime. $\square$

Theorem 3.15 ([30, pp. 149–150]). *Any prime number n that is not a Hurwitz prime can be written as a sum of four integer squares.*

Proof. Since we assume that n is not a Hurwitz prime, by Definition 3.6, we have $n = 0$, $\|n\| = 1$, or that n can be written as a product of non-unit Hurwitz integers. Since n is an prime number, $n \geq 2$ and thus $\|n\| \geq 2$. Hence, the first two cases are not possible. So $n = gh$ for some $g, h \in \mathbb{H}_{\text{Hurwitz}}$, with $\|g\|, \|h\| \neq 1$. Let $h = a + bi + cj + dk$. Then we have

$$n = g(a + bi + cj + dk). \quad (1)$$

Note that $\bar{n} = \overline{g(a + bi + cj + dk)} = \overline{(a + bi + cj + dk)} \bar{g} = (a - bi - cj - dk) \bar{g}$. Since $n \in \mathbb{R}$, we have $\bar{n} = n$. Hence,

$$n = (a - bi - cj - dk) \bar{g}. \quad (2)$$

Multiplying (1) and (2), we get $n^2 = g(a + bi + cj + dk)(a - bi - cj - dk) \bar{g} = g(a^2 + b^2 + c^2 + d^2) \bar{g}$. Since $a^2 + b^2 + c^2 + d^2 \in \mathbb{R}$, we have $n^2 = (a^2 + b^2 + c^2 + d^2) g \bar{g} = (a^2 + b^2 + c^2 + d^2) \|g\|^2$. Also, we have $a^2 + b^2 + c^2 + d^2 = \|h\|^2$, so $n^2 = \|h\|^2 \|g\|^2$. Since $g, h \in \mathbb{H}_{\text{Hurwitz}}$, by Theorem 3.3, we have $\|g\|^2, \|h\|^2 \in \mathbb{Z}$. Note that the only ways to factorise n^2 are $1 \times n^2$, $n^2 \times 1$, and $n \times n$. Since $\|g\|, \|h\| \neq 1$, we have $\|g\|^2, \|h\|^2 \neq 1$. Therefore, we have $\|g\|^2 = \|h\|^2 = n$. So

$$a^2 + b^2 + c^2 + d^2 = \|h\|^2 = n. \quad (3)$$

Since $h \in \mathbb{H}_{\text{Hurwitz}}$, either $a, b, c, d \in \mathbb{Z}$ or $a, b, c, d \in \mathbb{Z} + \frac{1}{2}$. If $a, b, c, d \in \mathbb{Z}$, then (3) expresses n as a sum of four integer squares. Thus, we assume

$a, b, c, d \in \mathbb{Z} + \frac{1}{2}$. Next, we define a quaternion ω depending on whether each of a, b, c , and d is $2n + \frac{1}{2}$ or $(2n + 1) + \frac{1}{2}$, for some $n \in \mathbb{Z}$. The motivation of this definition is to write

$$h = \omega + A + Bi + Cj + Dk, \quad (4)$$

where ω is a unit quaternion and A, B, C, D are even integers. The definition of ω for each case can be found in Table 1.

a	b	c	d	ω
$2n + 1/2$	$2n + 1/2$	$2n + 1/2$	$2n + 1/2$	$(1 + i + j + k)/2$
$2n + 1/2$	$2n + 1/2$	$2n + 1/2$	$(2n + 1) + 1/2$	$(1 + i + j - k)/2$
$2n + 1/2$	$2n + 1/2$	$(2n + 1) + 1/2$	$2n + 1/2$	$(1 + i - j + k)/2$
$2n + 1/2$	$2n + 1/2$	$(2n + 1) + 1/2$	$(2n + 1) + 1/2$	$(1 + i - j - k)/2$
$2n + 1/2$	$(2n + 1) + 1/2$	$2n + 1/2$	$2n + 1/2$	$(1 - i + j + k)/2$
$2n + 1/2$	$(2n + 1) + 1/2$	$2n + 1/2$	$(2n + 1) + 1/2$	$(1 - i + j - k)/2$
$2n + 1/2$	$(2n + 1) + 1/2$	$(2n + 1) + 1/2$	$2n + 1/2$	$(1 - i - j + k)/2$
$2n + 1/2$	$(2n + 1) + 1/2$	$(2n + 1) + 1/2$	$(2n + 1) + 1/2$	$(1 - i - j - k)/2$
$(2n + 1) + 1/2$	$2n + 1/2$	$2n + 1/2$	$2n + 1/2$	$(-1 + i + j + k)/2$
$(2n + 1) + 1/2$	$2n + 1/2$	$2n + 1/2$	$(2n + 1) + 1/2$	$(-1 + i + j - k)/2$
$(2n + 1) + 1/2$	$2n + 1/2$	$(2n + 1) + 1/2$	$2n + 1/2$	$(-1 + i - j + k)/2$
$(2n + 1) + 1/2$	$2n + 1/2$	$(2n + 1) + 1/2$	$(2n + 1) + 1/2$	$(-1 + i - j - k)/2$
$(2n + 1) + 1/2$	$(2n + 1) + 1/2$	$2n + 1/2$	$2n + 1/2$	$(-1 - i + j + k)/2$
$(2n + 1) + 1/2$	$(2n + 1) + 1/2$	$2n + 1/2$	$(2n + 1) + 1/2$	$(-1 - i + j - k)/2$
$(2n + 1) + 1/2$	$(2n + 1) + 1/2$	$(2n + 1) + 1/2$	$2n + 1/2$	$(-1 - i - j + k)/2$
$(2n + 1) + 1/2$	$(2n + 1) + 1/2$	$(2n + 1) + 1/2$	$(2n + 1) + 1/2$	$(-1 - i - j - k)/2$

Table 1: Definition of ω in each case.

We can verify that $\|\omega\| = 1$ in each case. Also, by comparing the constituents in (4), we can verify that A, B, C, D are even integers. Hence, from (3) and (4), we have

$$\begin{aligned}
n &= \|h\|^2 = h\bar{h} \\
&= (\omega + A + Bi + Cj + Dk) (\overline{\omega + A + Bi + Cj + Dk}) \\
&= (\omega + A + Bi + Cj + Dk) \|\omega\|^2 (\overline{\omega + A + Bi + Cj + Dk}) \\
&= (\omega + A + Bi + Cj + Dk) (\bar{\omega}\omega) (\overline{\omega + A + Bi + Cj + Dk}) \\
&= [(\omega + A + Bi + Cj + Dk)\bar{\omega}] [\omega (\overline{\omega + A + Bi + Cj + Dk})].
\end{aligned}$$

Since A, B, C, D are even, there exist $m_A, m_B, m_C, m_D \in \mathbb{Z}$ such that $A = 2m_A$, $B = 2m_B$, $C = 2m_C$, and $D = 2m_D$. Notice that $(\omega + A + Bi + Cj +$

$Dk)\bar{\omega}$ is a quaternion where all constituents are integers. For example, in the case $\omega = (1 + i + j + k)/2$,

$$\begin{aligned}
(\omega + A + Bi + Cj + Dk)\bar{\omega} &= \omega\bar{\omega} + 2m_A\bar{\omega} + 2m_B\bar{\omega} + 2m_C\bar{\omega} + 2m_D\bar{\omega} \\
&= \|\omega\| + 2m_A \left(\frac{1 - i - j - k}{2} \right) + 2m_B \left(\frac{1 - i - j - k}{2} \right) \\
&\quad + 2m_C \left(\frac{1 - i - j - k}{2} \right) + 2m_D \left(\frac{1 - i - j - k}{2} \right) \\
&= 1 + m_A(1 - i - j - k) + m_B(1 - i - j - k) \\
&\quad + m_C(1 - i - j - k) + m_D(1 - i - j - k).
\end{aligned}$$

We can prove other cases using the same approach. Therefore, we can find $u, v, w, x \in \mathbb{Z}$ satisfying $(\omega + A + Bi + Cj + Dk)\bar{\omega} = u + vi + wj + xk$. Since

$$\begin{aligned}
\overline{(\omega + A + Bi + Cj + Dk)\bar{\omega}} &= \bar{\bar{\omega}} \overline{(\omega + A + Bi + Cj + Dk)} \\
&= \omega \overline{(\omega + A + Bi + Cj + Dk)},
\end{aligned}$$

we have $\omega \overline{(\omega + A + Bi + Cj + Dk)} = u - vi - wj - xk$. Therefore,

$$n = (u + vi + wj + xk)(u - vi - wj - xk) = u^2 + v^2 + w^2 + x^2,$$

which is a sum of four integer squares. □

In Lemma [3.14](#), we prove that no odd prime is a Hurwitz prime. Therefore, with Theorem [3.15](#), it is sufficient to conclude that every odd prime is a sum of four integer squares. Furthermore, the only even prime 2 is a sum of four integer squares since $2 = 1^2 + 1^2 + 0^2 + 0^2$. Hence, every prime number can be written as a sum of four squares. By Corollary [3.5](#), all natural numbers are sums of four squares, completing the proof of Lagrange's four-square theorem. This theorem has applications in many areas of mathematics. An example is quantum computing in [\[24\]](#).

4 Rotations in $\mathbb{R}^3$

In this section, we first introduce the concept of Euler angles and explain why it leads to the problem of gimbal lock. Then, we discuss the connection between quaternions and rotations. Finally, we explain why quaternion is a better way to describe a rotation, compared to Euler angle.

4.1 Euler angles and gimbal lock

Euler angles are a method to parameterise rotations in $\mathbb{R}^3$. It is based on the fact that any rotation in $\mathbb{R}^3$ can be written as the composition of three rotations, about three mutually orthogonal axes.

For example, using the standard basis, we can write any rotation as the composition of a rotation about the z -axis, a rotation about the x -axis, and finally a rotation about the y -axis. Any order of the axes is possible, but we will use the order stated in the above example, which is the same as in [8, Section 2.8]. We use $R(\theta, \phi, \psi)$ to denote a rotation of ψ in the z -axis, followed by a rotation of θ in the x -axis and finally a rotation of ϕ in the y -axis. Therefore, we have the following theorem.

Theorem 4.1. *Let $\theta, \psi \in (-\pi, \pi]$, $\phi \in [-\frac{\pi}{2}, \frac{\pi}{2}]$. Then we have*

$$R(\theta, \phi, \psi) = R_x(\theta)R_y(\phi)R_z(\psi),$$

where $R_x(\theta)$, $R_y(\phi)$ and $R_z(\psi)$ are matrices representing rotations of angles θ , ϕ and ψ about the x -, y - and z -axes respectively.

However, Euler angles can sometimes cause problems, such as gimbal lock. We illustrate this problem using a similar example as in [9, p. 13]. It occurs when two of the three gimbal axes align, restricting the direction of rotations it can represent. For instance, this happens when $\phi = \pi/2$.

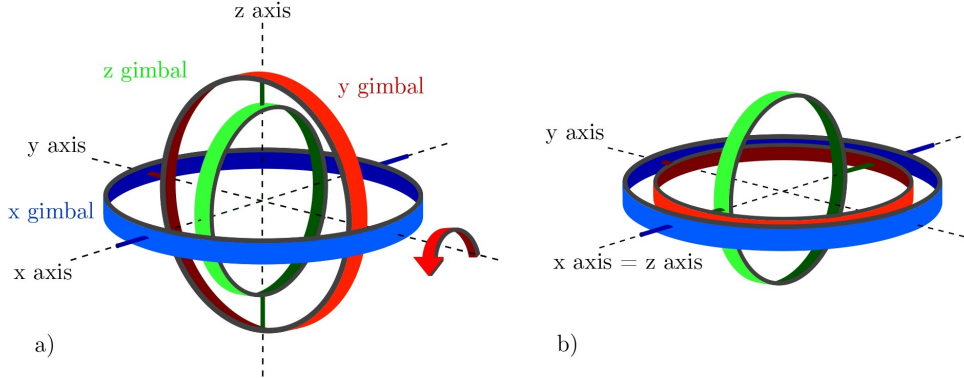


Figure 1: Occurrence of gimbal lock. [34, Figure 9]

Figure 1(a) shows the initial position of the three axes, which are orthogonal to each other. At this stage, we can rotate in any direction. However, after a rotation of $\pi/2$ along the y -axis, the x - and z -gimbals coincide, as

illustrated in Figure 1 (b). Therefore, there are only two orthogonal axes and thus it cannot represent rotations in every direction.

Using $R_x(\theta) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta \\ 0 & \sin \theta & \cos \theta \end{pmatrix}$, $R_y(\phi) = \begin{pmatrix} \cos \phi & 0 & \sin \phi \\ 0 & 1 & 0 \\ -\sin \phi & 0 & \cos \phi \end{pmatrix}$ and $R_z(\psi) = \begin{pmatrix} \cos \psi & -\sin \psi & 0 \\ \sin \psi & \cos \psi & 0 \\ 0 & 0 & 1 \end{pmatrix}$, which are stated in [8, pp. 46–47], we can show that

$$R(\theta, \pi/2, \psi) = \begin{pmatrix} 0 & 0 & 1 \\ \sin(\psi + \theta) & \cos(\psi + \theta) & 0 \\ -\cos(\psi + \theta) & \sin(\psi + \theta) & 0 \end{pmatrix}.$$

This matrix has non-unique choices of θ and ψ . For example, for any $t \in \mathbb{R}$, the rotation given by $\theta = t$ and $\psi = 0$ is the same as that given by $\theta = 0$ and $\psi = t$, since

$$R(t, \pi/2, 0) = \begin{pmatrix} 0 & 0 & 1 \\ \sin t & \cos t & 0 \\ -\cos t & \sin t & 0 \end{pmatrix} = R(0, \pi/2, t).$$

This can cause issues when attempting to make smooth changes in orientation near the point of degeneracy, as a minor adjustment can require a large, discontinuous change in another. In applications such as aerospace navigation or animation, this can result in sudden jumps that may be unnatural or pose a safety risk as explained in [15, § IV].

4.2 Rotations using quaternions

In this subsection, we introduce the representation of a rotation in $\mathbb{R}^3$ by the conjugation of a unit quaternion, with reference to [32]. This method can avoid the problem of gimbal lock. However, we first need to prove that given a vector $v \in \mathbb{R}^3$, its conjugation with a unit quaternion q , which results in qvq^{-1} , is also a vector in $\mathbb{R}^3$.

Theorem 4.2. *Let $q \in \mathbb{H}^1$ and $v \in \mathbb{R}^3$. Then $qvq^{-1} \in \mathbb{R}^3$.*

Proof. Let $q = a + bi + cj + dk$ and $v = \alpha i + \beta j + \gamma k$. Then, $q^{-1} = \frac{1}{\|q\|^2}(a - bi - cj - dk) = (a - bi - cj - dk)$. Hence, $qvq^{-1} = (a + bi + cj + dk)(\alpha i + \beta j + \gamma k)(a - bi - cj - dk)$. So the real part of qvq^{-1} is

$$\begin{aligned} & (a\alpha b + a\beta c + a\gamma d - b\alpha a + b\beta d - b\gamma c \\ & - c\alpha d - c\beta a + c\gamma b + d\alpha c - d\beta b - d\gamma a) \\ & = 0. \end{aligned}$$

Therefore, $qvq^{-1} \in \mathbb{R}^3$. □

If we fix $q \in \mathbb{H}^1$, then we can define a linear transformation $T_q : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ by $T_q(v) = qvq^{-1}$ for any $v \in \mathbb{R}^3$. Using the standard basis of $\mathbb{R}^3$, namely $\left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \right\}$, the matrix representing this linear transformation is

$$\begin{pmatrix} a^2 + b^2 - c^2 - d^2 & 2bc - 2ad & 2ac + 2bd \\ 2bc + 2ad & a^2 - b^2 + c^2 - d^2 & -2ab + 2cd \\ -2ac + 2bd & 2ab + 2cd & a^2 - b^2 - c^2 + d^2 \end{pmatrix}.$$

We denote this matrix by M_q . This matrix can be obtained by finding the images of each basis vectors, because each column of the matrix is the coordinate vector of the image of the basis vector. A detailed derivation of this matrix can be found in [11, Chapter 15.3]. Next, we define the special orthogonal group, which allows us to establish some properties of M_q .

Definition 4.3 ([23, p. 223]). The special orthogonal group, $\text{SO}(n)$, is the set of all $n \times n$ orthogonal matrices with determinant 1, with respect to the standard inner product of $\mathbb{R}^n$, namely $\langle u, v \rangle = \sum_{i=1}^n u_i v_i$ for any $u = (u_1, \dots, u_n)^T$ and $v = (v_1, \dots, v_n)^T$ in $\mathbb{R}^n$. The binary operation is matrix multiplication.

We can calculate the inner products of the columns in M_q , showing that the inner product of any two distinct columns is 0, and the inner product of any column with itself is 1. This verifies that M_q is orthogonal. We can also calculate $\det(M_q)$ and show that it is 1. Hence, we can conclude that $M_q \in \text{SO}(3)$. This leads to the following theorem.

Theorem 4.4 ([7, p. 24]). *The function $\Phi : \mathbb{H}^1 \rightarrow \text{SO}(3)$ defined by $\Phi(q) = M_q$ for all $q \in \mathbb{H}^1$ is a surjective group homomorphism.*

Recall that elements in $\text{SO}(3)$ are rotations in $\mathbb{R}^3$. Therefore, the function Φ in Theorem 4.4 being surjective implies that any rotation in $\mathbb{R}^3$ can be written as a unit quaternion.

Quaternion rotation, as mentioned before, avoids the issue of gimbal lock and allows rotations to be represented as smooth compositions using spherical linear interpolation (slerp). This method provides constant speed, smooth, and intermediate angles between two orientations on the shortest path between two points, addressing the interpolation issues commonly encountered with Euler angles as shown in [28]. Because of this, quaternions are generally considered more computationally efficient and numerically stable

than Euler angles for representing and composing repeated 3D rotations, they are considered more computationally efficient due to using less operations with only and less memory as explained in more detail in [4, Sec. 2].

Example 4.5 (Rotation using a unit quaternion). Let $q = \frac{\sqrt{3}}{2} + \frac{\sqrt{2}}{4}(i + j)$ and $v = j + k$, where q represents a rotation by $\pi/3$ around the $\frac{1}{\sqrt{2}}(i + j)$ unit axis, and v is the point in $\mathbb{R}^3$ being rotated. Since q is a unit quaternion, its inverse is $q^{-1} = \bar{q} = \frac{\sqrt{3}}{2} - \frac{\sqrt{2}}{4}(i + j)$.

Denote the image of the rotation by v' . Then

$$\begin{aligned} v' &= qvq^{-1} = \left(\frac{\sqrt{3}}{2} + \frac{\sqrt{2}}{4}(i + j) \right) (j + k) \left(\frac{\sqrt{3}}{2} - \frac{\sqrt{2}}{4}(i + j) \right) \\ &= \left(-\frac{\sqrt{2}}{4} + \frac{\sqrt{2}}{4}i + \frac{2\sqrt{3} - \sqrt{2}}{4}j + \frac{2\sqrt{3} + \sqrt{2}}{4}k \right) \left(\frac{\sqrt{3}}{2} - \frac{\sqrt{2}}{4}(i + j) \right) \\ &= \frac{1 + \sqrt{6}}{4}i + \frac{3 - \sqrt{6}}{4}j + \frac{2 + \sqrt{6}}{4}k. \end{aligned}$$

Therefore, after a rotation by $\pi/3$ around the unit axis $\frac{1}{\sqrt{2}}(i + j)$, the point $j + k$, which corresponds to $(0, 1, 1)^T$, is mapped to $\frac{1+\sqrt{6}}{4}i + \frac{3-\sqrt{6}}{4}j + \frac{2+\sqrt{6}}{4}k$, which corresponds to $\left(\frac{1+\sqrt{6}}{4}, \frac{3-\sqrt{6}}{4}, \frac{2+\sqrt{6}}{4} \right)^T$.

5 Further extensions

The real numbers can be extended to the complex numbers, and the complex numbers can be extended to the quaternions. However, this is not the end of the chain of extensions.

5.1 Octonions

Complex numbers and quaternions are related to rotations in two- and four-dimensional spaces respectively. To study rotations in higher dimensions, Arthur Cayley constructed the octonions, the eight-dimensional numbers. We introduce the construction of octonions as described in [6, pp. 234–235]. Instead of using different symbols, we distinguish different imaginary parts by subscripts. So an octonion can be written as $a + bi_0 + ci_1 + di_2 + ei_3 + fi_4 + gi_5 + hi_6$, where $a, b, c, d, e, f, g, h \in \mathbb{R}$. The set of all octonions is denoted by $\mathbb{O}$. Two octonions can be added by summing the constituents of the real part and each imaginary unit respectively. Similar to the quaternions, to define the product of octonions, we first set up the rules for the products

of imaginary units and 1, which can be summarised in Table 2. Then, to multiply any two octonions, we expand the brackets and simplify it using these rules.

$\downarrow \times \rightarrow$	1	i_0	i_1	i_2	i_3	i_4	i_5	i_6
1	1	i_0	i_1	i_2	i_3	i_4	i_5	i_6
i_0	i_0	-1	i_2	$-i_1$	i_4	$-i_3$	$-i_6$	i_5
i_1	i_1	$-i_2$	-1	i_0	i_5	i_6	$-i_3$	$-i_4$
i_2	i_2	i_1	$-i_0$	-1	i_6	$-i_5$	i_4	$-i_3$
i_3	i_3	$-i_4$	$-i_5$	$-i_6$	-1	i_0	i_1	i_2
i_4	i_4	i_3	$-i_6$	i_5	$-i_0$	-1	$-i_2$	i_1
i_5	i_5	i_6	i_3	$-i_4$	$-i_1$	i_2	-1	$-i_0$
i_6	i_6	$-i_5$	i_4	i_3	$-i_2$	$-i_1$	i_0	-1

Table 2: Multiplication table for octonions. [2, Table 1]

From Table 2, we can see that multiplication of octonions is not commutative or associative. For example, $i_0 i_1 = i_2 \neq -i_2 = i_1 i_0$ and $(i_1 i_2) i_3 = i_0 i_3 = i_4 \neq -i_4 = i_1 i_6 = i_1 (i_2 i_3)$.

5.2 The Cayley-Dickson construction

The Cayley-Dickson Construction is a generalised procedure to define a sequence of $\mathbb{R}$ -algebras. We start with $\mathbb{R}$, and the first three iterations give us $\mathbb{C}$, $\mathbb{H}$, and $\mathbb{O}$. However, it defines a systematic way to continue constructing new numbers on this chain of extensions. We use the explanation in [12, p. 6].

Starting with the real numbers, we define complex numbers as a pair of real numbers (a, b) . Then we define addition of complex numbers as

$$(a, b) + (c, d) = (a + c, b + d), \quad (5)$$

multiplication as

$$(a, b)(c, d) = (ac - d\bar{b}, \bar{a}d + cb), \quad (6)$$

and conjugation as

$$\overline{(a, b)} = (\bar{a}, -b), \quad (7)$$

where the conjugation of a real number a is $\bar{a} = a$.

The next iteration of numbers is defined as a pair of numbers in the previous iteration, where addition, multiplication, and conjugation of numbers in this new iteration are defined as (5), (6), and (7) respectively. For example, following the Cayley-Dickson Construction process, quaternions can

be defined as a pair of complex numbers, and octonions can be defined as a pair of quaternions. The next number system formed is sedenions.

Throughout this construction process, some properties of the real numbers are lost. First, as we have seen in Section 2 and 5.1 respectively, the quaternions lose commutativity and the octonions lose associativity of multiplication. However, they retain the alternative property, which is $(xx)y = x(xy)$ and $y(xx) = (yx)x$ for any $x, y \in \mathbb{O}$, as stated in [27, p. 5]. This property is lost when we reach sedenions. The sedenions have zero divisors. For example, using the multiplication table of sedenions in [29, Figure 1], $(i_1 + i_{10})(i_5 - i_{14}) = 0$ and so

$$(i_1 + i_{10})[(i_5 - i_{14})i_4] = 2 \neq 0 = [(i_1 + i_{10})(i_5 - i_{14})]i_4.$$

Having zero divisors means that you cannot define a norm. If a norm $\|\cdot\|$ exists, then for any pair of zero divisors x and y , we have $\|xy\| = \|0\| = 0$ but $\|x\| \cdot \|y\| > 0$. This contradicts the requirement for a norm to be multiplicative. The lack of norm for sedenions is linked to the Hurwitz theorem.

Theorem 5.1 (Hurwitz theorem, [5, p. 1]). *Suppose there is an identity*

$$(x_1^2 + \cdots + x_n^2)(y_1^2 + \cdots + y_n^2) = z_1^2 + \cdots + z_n^2,$$

where $x_1, \dots, x_n, y_1, \dots, y_n \in \mathbb{C}$, and each z_k is of the form $\sum_{i=1}^n \sum_{j=1}^n c_{ij} x_i y_j$ with $c_{ij} \in \mathbb{C}$. Then $n = 1, 2, 4$, or 8 .

This implies that any finite-dimensional normed division $\mathbb{R}$ -algebra has dimension 1, 2, 4, or 8, explaining why Hamilton could not define a well-behaved multiplication of triplets as we discuss in Section 1. In other words, among all the $\mathbb{R}$ -algebras constructed in the Cayley-Dickson construction, the only normed division algebras are $\mathbb{R}$, $\mathbb{C}$, $\mathbb{H}$, and $\mathbb{O}$.

References

- [1] S. Axler. *Linear Algebra Done Right*. 4th ed. Undergraduate Texts in Mathematics. Cham: Springer, 2024. ISBN: 978-3-031-41026-0.
- [2] J. Baez. *Constructing the Octonions*. URL: <https://math.ucr.edu/home/baez/octonions/node3.html> (visited on 09/04/2026).
- [3] J. Baez and J. Huerta. ‘Division algebras and supersymmetry II’. In: *Advances in Theoretical and Mathematical Physics* 15.5 (2010). DOI: <https://doi.org/10.4310/ATMP.2011.v15.n5.a4>.

- [4] A. Cariow et al. ‘Structure and Principles of Operation of a Quaternion VLSI Multiplier’. In: *Applied Sciences* 14.18 (2024). DOI: [10.3390/app14188123](https://doi.org/10.3390/app14188123).
- [5] K. Conrad. *Hurwitz’s Theorem on Sums of Squares (by Linear Algebra)*. URL: <https://kconrad.math.uconn.edu/blurbs/linmultialg/hurwitzlinear.pdf> (visited on 20/04/2026).
- [6] J. H. Conway and R. K. Guy. *The book of numbers*. New York: Copernicus, 1996. ISBN: 978-0-387-97993-9.
- [7] J. H. Conway and D. A. Smith. *On Quaternions and Octonions: Their Geometry, Arithmetic, and Symmetry*. A K Peters, 2002. ISBN: 1-56881-134-9.
- [8] J. J. Craig. *Introduction to Robotics: Mechanics and Control*. 3rd ed. Upper Saddle River: Pearson, 2005. ISBN: 0-13-123629-6.
- [9] J. Diebel. *Representing Attitude: Euler Angles, Unit Quaternions, and Rotation Vectors*. URL: https://www.astro.rug.nl/software/kapteyn/_downloads/fd5693488984818bb79c2eedad397482/attitude.pdf (visited on 20/04/2026).
- [10] R. Duram and M. Güzeltepe. ‘Encoder Hurwitz Integers: The Hurwitz integers that have the ”division with small division” property’. In: (2021). DOI: <https://doi.org/10.21203/rs.3.rs-1053551/v2>.
- [11] J. Gallier and J. Quaintance. *Linear Algebra for Computer Vision, Robotics, and Machine Learning*. URL: <https://personalpages.bradley.edu/~smiah/files/webBooks/linalg-I-f.pdf> (visited on 15/04/2026).
- [12] R. Gardner. *Supplement. The Cayley-Dickson Construction and Nonassociative Algebras*. URL: <https://faculty.etsu.edu/gardnerr/4127/notes-Herstein/notes-Cayley-Dickson.pdf> (visited on 15/04/2026).
- [13] A. S. Garge. ‘Tracing the History of a Theorem: Lagrange’s Four Squares Theorem’. In: *At Right Angles* 1.2 (2012), pp. 5–9. ISSN: 2582-1873. URL: <http://publications.azimpremjiuniversity.edu.in/id/eprint/1581>.
- [14] L. Goldmakher. *A Brief Introduction to the Cauchy-Schwarz and Hölder Inequalities*. URL: <https://web.williams.edu/Mathematics/lg5/Cauchy-Schwarz.pdf> (visited on 28/04/2026).

- [15] F. R. González Díaz et al. ‘A Quaternionic Approach to Teaching 3D Rotations and the Resolution of Gimbal Lock’. In: *arXiv preprint arXiv:2511.04452* (2025). DOI: [10.48550/arXiv.2511.04452](https://doi.org/10.48550/arXiv.2511.04452).
- [16] W. R. Hamilton. *On Quaternions*. Ed. by D. R. Wilkins. URL: <https://www.maths.tcd.ie/pub/HistMath/People/Hamilton/Quatern2/Quatern2.pdf> (visited on 13/04/2026).
- [17] W. R. Hamilton. ‘Theory of Conjugate Functions, or Algebraic Couples; with a Preliminary and Elementary Essay of Algebra as the Science of Pure Time’. In: *The Transactions of the Royal Irish Academy* 17 (1837), pp. 293–422.
- [18] W. R. Hamilton. *On Quaternions, or on a New System of Imaginaries in Algebra*. Ed. by D. R. Wilkins. URL: <https://www.maths.tcd.ie/pub/HistMath/People/Hamilton/OnQuat/OnQuat.pdf> (visited on 14/04/2026).
- [19] G. H. Hardy and E. M. Wright. *An Introduction to the Theory of Numbers*. Ed. by D. R. Heath-Brown, J. Silverman and A. Wiles. 6th ed. Oxford University Press, 2008. ISBN: 978-0-199-21986-5.
- [20] B. K. P. Horn. ‘Quaternions: what are they, and why do we need to know?’ In: *Acta Crystallographica Section A Foundations and Advances* (2020), pp. 556–558. DOI: <https://doi.org/10.1107/S2053273320010359>.
- [21] D. Khattar and N. Agrawal. *Group Theory*. Cham: Springer, 2023. ISBN: 978-3-031-21307-6.
- [22] D. Khattar and N. Agrawal. *Ring Theory*. Cham: Springer, 2023. ISBN: 978-3-031-29440-2.
- [23] M. R. Kibler. *Galois Fields and Galois Rings Made Easy*. Elsevier, 2017. ISBN: 978-1-785-48235-9.
- [24] J. Lacalle and L. N. Gatti. ‘Discrete quantum computation and Lagrange’s four-square theorem’. In: *Quantum information processing* 19.34 (2020). DOI: <https://doi.org/10.1007/s11128-019-2528-7>.
- [25] M. Lackmann. ‘The octonionic projective plane’. In: *2019-20 MATRIX Annals*. Ed. by D. R. Wood et al. Cham: Springer, 2021. ISBN: 978-3-030-62497-2.
- [26] M. Pan and W. Zhang. ‘Quotients of Hurwitz Primes’. In: (2019). DOI: <https://doi.org/10.48550/arXiv.1904.08002>.
- [27] R. D. Schafer. *An Introduction to Nonassociative Algebras*. New York: Academic Press, 1966.

- [28] K. Shoemake. ‘Animating Rotation with Quaternion Curves’. In: *Proceedings of the 12th Annual Conference on Computer Graphics and Interactive Techniques (SIGGRAPH ’85)*. New York, NY, USA: Association for Computing Machinery, 1985, pp. 245–254. ISBN: 0-89791-166-0. DOI: [10.1145/325334.325242](https://doi.org/10.1145/325334.325242). URL: <https://dl.acm.org/doi/pdf/10.1145/325334.325242>.
- [29] Y. Soykan. ‘Tribonacci and Tribonacci-Lucas Sedenions’. In: *Mathematics* 7.1 (2019). DOI: <https://doi.org/10.3390/math7010074>.
- [30] J. Stillwell. *Elements of Number Theory*. Undergraduate Texts in Mathematics. Springer, 2003. ISBN: 978-0-387-95587-2.
- [31] J. Suranyi and P. Erdős. *Topics in the Theory of Numbers*. Undergraduate Texts in Mathematics. Springer Science Business Media, 2003. ISBN: 978-0-387-95320-5.
- [32] T. Tao. *Quaternions and spherical trigonometry*. URL: <https://terrytao.wordpress.com/2024/12/19/quaternions-and-spherical-trigonometry/> (visited on 29/04/2026).
- [33] B. L. van der Waerden. ‘Hamilton’s Discovery of Quaternions’. In: *Mathematics Magazine* 49.5 (1976), pp. 227–234.
- [34] J. Zeitlhöfler. *Nominal and observation-based attitude realization for precise orbit determination of the Jason satellites*. URL: <https://mediatum.ub.tum.de/doc/1535899/1535899.pdf> (visited on 18/04/2026).